

The Revolutionary Theory of Quantum Mechanics

Introduction

You have probably heard of the word “quantum.” But what even is a “quantum,” and how does it relate to quantum mechanics? What even is quantum mechanics? Well, quantum mechanics is the explanation of what happens to particles at the atomic and subatomic levels, and its impacts on physics fundamentally changed our modern understanding of the world that makes us up. In this research paper, I will explain in detail the properties of quantum mechanics, how it came into existence from some of the greatest minds in history, and how it shows its amazing applications in our modern world.

What is quantum mechanics?

Properties of Quantum Particles: Measurement matters

Quantum mechanics has a few rules about what happens to particles at the smallest levels of reality. These include wave-particle duality, superposition, and entanglement.¹

Wave-Particle Duality: It’s a particle... but it’s a wave

For most of human history, we thought that light was a natural part of our world. But multiple scientists centuries ago thought about this as a problem and discovered that light is actually a particle shooting through the air into your eyes. Then, the double-slit experiment² (shooting a particle through slits to show that it always landed on the other side as a wave) proved that these light “particles” are actually waves that collide to form particles of light. It turns out that, according to quantum mechanics, the answer is neither and both at the same time- light is a wave when you aren’t measuring it, and light is a particle when you are. Although this sounds counterintuitive, quantum mechanics has equations to show this property for not just light (photons) but also for electrons and many other subatomic particles.

¹ *What Is Quantum Physics?* (2026). Caltech Science Exchange. [What Is Quantum Physics?](#)

² admin. (2025, July 28). *Famous double-slit experiment holds up when stripped to its quantum essentials*. MIT Physics. [Famous double-slit experiment holds up when stripped to its quantum essentials » MIT Physics](#)

Superposition: Spinning both ways

Quantum particles also are notorious for being in multiple states at the same time, the most recognizable of these being spin. When you measure or observe a particle, it either spins clockwise or counterclockwise, but in physics these are called “up” or “down” spins. In quantum superposition, however, both of these spins can be present at the same time before measurement of a particle. An significant instance of this property is Shrödinger’s Cat, in which the 50/50 chance of the result of the measurement of a superimposed particle results in either a cat being alive or dead. To the human who cannot currently see the cat yet, the cat is both alive and dead at the same time- just like how the measured particle is both spinning up and down at the same time until a human checks.

Quantum Entanglement: Information goes the distance

Quantum entanglement is what happens when two particles, sometimes billions of light-years apart, form a connected system based solely on the information available about those two particles. Whenever you measure a certain property of one particle in the system, the other particle already has the opposite of the information that applied to the original particle. Also, the complete state of one particle cannot be determined from just the information about the one particle; you need to know about both entangled particles to be able to completely know about one particle. Again, this sounds counterintuitive, but subatomic particles can behave quite differently from what we experience in everyday life- including quantum entanglement.

Probabilistic Nature of Particles: Everywhere at once

Quantum mechanics is hard to visualize because it doesn’t describe the world that we see, so mathematics comes in as a useful way to form equations that explain most of the main principles of quantum mechanics. One of these equations is called the Shrödinger equation, which explains the probabilities of particles appearing at a certain point in space and time. These probabilities can be graphed as a wave function, which explains how likely there is to be a particle there when you collapse the wave function (a fancier way to say measuring or observing the particle.) You may have heard of electron clouds as being a more accurate model of the atom as opposed to simply orbits around the nucleus. In fact, this is just the 3D version of the electron probability waves that result from the Shrödinger equation, and are very useful for determining certain properties of atoms.

How did quantum mechanics come to be?

The Problem: Classical physics doesn't work anymore

In the late 1800s, physicists were coming up with problems. Classical physics, which was used to describe how things acted in the world through laws like Newton's, seemed to be useless when trying to describe the results of experiments for atomic and subatomic particles (most of which were just being discovered: for example, 1897 is when the electron was found.) Most scientists dismissed this concern by simply creating an exception to the rules of classical physics and called it a day. One of these scientists was named Max Planck.

Max Planck: The colorful discovery of the “quantum”

In 1900, Max Planck was working on an experiment about black-body radiation and was trying to explain why perfectly absorbent objects still emitted heat and light.³ According to classical physics, only heated objects should emit an infinite array of ultraviolet light, which of course does not make any sense to what we see in today's world. Max Planck agreed, and came up with a theory that a finite length called the “quantum” (later named Planck's constant) defined the wavelength of arrays of light. He proposed this theory in December and won the Nobel Prize, but not until 1918- after all, this theory was impossible with classical physics alone. However, Planck's theory was helped up by a certain patent clerk in 1905 named Albert Einstein.

Einstein and Bohr: The great debate over locality

Albert Einstein produced four famous research papers in 1905, but only two were about general relativity. His favorite paper was actually on the photoelectric effect, an article explaining why certain metals emit electrons with varying properties based on the color of light being emitted at it. Einstein explained that the only way that the color of light could vary based on its metal electron output is by having different wavelengths, and his main formula for these wavelengths includes using the “quantum” that Max Planck came up with. Next came physicist Neils Bohr, who used Max Planck's “quantum” to come up with a more accurate model for the atom that wouldn't fall apart. Bohr was visited by a student, Werner Heisenberg, who thought with Bohr about how to form a new set of physical laws from the ground up (matrix mechanics, or the observable part of quantum mechanics). Erwin Shrödinger added on by formulating equations for waves, or the part of quantum mechanics that isn't observable. By this point, two sides of quantum mechanics formed: one with Bohr and Heisenberg, which relied on the instant transformation of information and a more probabilistic nature of particles, and one with Einstein and Shrödinger on non-instant transfer of information and a less probabilistic nature of particles. Einstein famously said in a letter to Bohr: “God does not play dice with the universe,” implying that not everything in quantum mechanics can be random and everywhere at once because that contradicted his general theory of relativity. Einstein also debated the “locality principle:” that no information or effect can travel faster than the speed of light, meaning that entangled particles

³ Ball, Phillip. “The Tumultuous Birth of Quantum Mechanics.” *Physics*, vol. 18, 4 Feb. 2025, physics.aps.org/articles/v18/24.

must have been closer together before they were spread apart to allow the share of information. Overall, Bohr published a confusing paper that basically said “No, Einstein is lying” in scientific language to prove that Einstein wasn’t actually right, and he ended up winning the debate. Except... modern quantum technology has now proven that Einstein was right and that information cannot travel faster than the speed of light.⁴

What are the impacts of quantum mechanics in our modern world?

Quantum technologies

In our modern world, it may seem that quantum mechanics is useless- after all, it describes how tiny particles work. But subatomic particles have successfully made many useful technologies over the years, and with quantum mechanics these technologies have the potential to become many orders of magnitude more efficient.

Quantum computing: Data to no limit

Today, all sorts of technological advancements have been made via quantum mechanics.⁵ Your GPS? An atomic clock stabilizing your place and time. Your nearest solar panels? The photovoltaic effect in action. An MRI machine? Using the quantum spin of particles to focus a radio wave through the human body, generating near-perfect imaging. Even a laser stands for “Light Amplification by Stimulated Emission of Radiation,” which uses quantum atoms to focus radiation (not the deadly kind, ionizing radiation, but simply light irradiating) onto a single, bright red point. In the future, scientists may develop a quantum computer (despite computers already running on some form of quantum technology today,) which keeps all its bits of data in the form of superimposed particles. Basically, instead of data being 010110, data is superimposed so that all bits will be 0s and 1s at the same time. This will increase its data capacity by orders of magnitude if quantum computing ever becomes popular.

Quantum outside of computing: From encryption to genes to energy

Quantum technologies can improve the future, but not just through computing. Quantum encryption will allow passwords and IDs to be kept safer, quantum simulations will instruct

⁴ Franson, J., & Potocki, K. (n.d.). *AN EXPERIMENTAL TEST OF THE QUANTUM THEORY*. [AN EXPERIMENTAL TEST OF THE QUANTUM THEORY](#)

⁵ *Quantum applications today* | Institute for Quantum Computing | University of Waterloo. (2026, January 5). Uwaterloo.ca. [Quantum applications today | Institute for Quantum Computing | University of Waterloo](#)

scientists on how to purify food and water from certain chemicals, and quantum computing will maximize energy output from power plants as well as perfecting the most resistant gene for farm crops.⁶ Even still, as quantum mechanics encounters more and more physics scientists and breakthroughs, the future of quantum mechanics is uncertain. It could allow for technologies that boost daily life and make near-perfect digital commands, but it could also allow for technologies that could destroy the world itself like atom bombs. Either way, the sheer power from quantum mechanics has so far created some of the world's best inventions, of which we rely on today and will into the future.

Conclusion

If you still have questions after reading this and don't fully understand quantum mechanics, good job! After all, a famous quote said by Nobel laureate physicist Richard Feynman is that "I think I can safely say that nobody understands quantum mechanics." When Mr. Feynman first said this quote at a talk in 1964 at Cornell University, most people were in disbelief of how a Nobel laureate physicist said they didn't understand quantum mechanics. But Mr. Feynman was a smart man, and he knew what he (most likely, he isn't alive today so I can't confirm) meant by that quote: that quantum mechanics is not just a set of laws, but a never ending pursuit of knowledge for the world around us that makes us up. That pursuit of knowledge is what enabled all of the famous scientists from Planck to Feynman and beyond to create a theory, invention, or idea that changed the world. Without the scientific branch of quantum mechanics, there would be no GPS, no computers, and our world full of technological advancements would cease to exist. So, remember to thank the scientists that put their mind to creating one of the most confusing yet amazing scientific revolutions ever: quantum mechanics.

⁶ "Quantum Physics to Everyday Life – Shaping Society's Future | CERN & Society Foundation." *Cernandsocietyfoundation.cern*, 2025, cernandsocietyfoundation.cern/news/quantum-physics-everyday-life-shaping-societys-future.